

Adaptive fuzzy fixed-time control for stochastic nonlinear systems with input saturation

Measurement and Control

2026, Vol. 59(6) 826–841

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DOI: 10.1177/00202940251355351

journals.sagepub.com/home/mac



Liandi Fang¹ , Zhenzhen Long² and Daohong Zhu¹

Abstract

This essay investigates the issue of adaptive fuzzy fixed-time control for a type of stochastic nonlinear systems, inputs of which are perturbed by saturation. First of all, an improved practical fixed-time stable criterion is presented for stochastic nonlinear system. Then, Gaussian error function (erf) and fuzzy logic systems (FLSs) are successively unitized to handle the issue of input saturation and unknown functions for the considered systems. Based on the given criterion and adding a power integral method, an adaptive fuzzy fixed-time controller is designed, which can ensure that all the variables converge to a compact set containing equilibrium point within a fixed time. Eventually, the simulation result of the numerical example and the one-link manipulator system demonstrate the validity of the suggested control strategy.

Keywords

fixed-time control, input saturation, stochastic nonlinear systems, fuzzy logic system

Date received: 7 February 2025; accepted: 2 June 2025

Introduction

There is no doubt that the effective controller design and stability analysis for nonlinear systems are always hot topics in the field of control.^{1–4} Due to its higher accuracy, faster convergence speed and strong disturbance rejection performance, the finite-time control strategies have garnered significant attention.^{5–7} As a special form of finite-time controllers, the fixed-time stabilizer has been extensively studied in recent years. Unlike finite-control, the estimation of the settling time in fixed-time control is independent of the initial state. This characteristic makes it particularly advantageous in application. For instance, reference⁸ has developed a fixed-time stability criteria for linear systems, which has provided the theoretical foundation for fixed-time control. In reference,⁹ a new fixed-time control program has been developed for time-delay systems to reach global synchronization within fixed time. Reference¹⁰ has introduced a fixed-time fuzzy control method for nonlinear system. Additionally, reference¹¹ has presented an innovative fixed-time constrained control protocol that addressed the control issue of high-order sliding mode system with asymmetric output constraints. The adaptive fixed-time control scheme has been proposed for a class of MIMO nonstrict feedback system in reference.¹² It should be pointed out that, to alleviate the dependence of the settling time on design parameters, the predefined-time/specified-time stability is recently

reported in reference,¹³ in which the least upper bound of the settling time can be preset irrespective of initial values and any other design parameters. The predefined-time/specified-time control which can ensure states converge to zero (or a residual set) within a specified finite time interval, has attracted much attention. Reference¹⁴ considered specified-time convergent multi-agent system for distributed optimization with a time-varying objective function. Reference¹⁵ has explored practical prescribed-time tracking control with bounded time-varying gain under nonvanishing uncertainties, and prescribed-time input-to-state stabilization has been addressed for nonlinear systems by using bounded time-varying feedback in reference.¹⁶

However, these results primarily focused on deterministic systems, while many practical engineering systems are often subject to stochastic factors, such as permanent synchronous motor¹⁷ and rigid-flexible wing system,¹⁸ etc. Therefore, the fixed-time control of stochastic nonlinear systems is broader practical

¹College of Mathematics and Computer Science, Tongling University, Tongling, Anhui, China

²School of Mathematics and Statistics, Anhui Normal University, Wuhu, Anhui, China

Corresponding author:

Daohong Zhu, College of Mathematics and Computer Science, Tongling University, Tongling, Anhui 244061, China.

Email: 2022096@tlu.edu.cn



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significance. Recently, the notion of fixed-time stability in probability and Lyapunov criterion for stochastic nonlinear system have been offered by Yu et al.¹⁹ Despite the important theoretical and practical value of fixed-time control for stochastic nonlinear systems, research achievement in this area remain limited. In recent years, scholars have begun to explore this field. References^{20,21} addressed the fixed-time stability issues for switched stochastic systems and multi-agent stochastic systems, respectively. In addition, references^{22,23} have improved the fixed-time Lyapunov stable criterion in reference,¹⁹ thereby providing more theoretical tools for fixed-time stability of stochastic systems.

In practical engineering applications, nonlinear systems are widely prevalent in fields such as aerospace, robotics, and power systems. However, these systems are often subject to various uncertain factors, including external disturbances, parameter perturbations, which pose significant challenges to the design of controllers.^{10,24–26} To address these uncertainties, scholars have proposed a variety of control strategies. References^{27,28} have successively investigated the adaptive fuzzy state-feedback and output-feedback fault-tolerant control for MIMO nonlinear systems. Reference²⁹ has proposed an adaptive fuzzy finite-time tracking control method to effectively address the tracking problem for nonlinear systems. Reference³⁰ future proposed a fuzzy semi-globally finite-time control method, significantly enhancing the tracking performance of systems. Fuzzy tracking control for SISO nonlinear systems with unknown directions and MIMO nonlinear systems with actuator and sensor failures have been considered in references.^{31,32} In addition, references^{33,34} have constructed the adaptive event-triggered controllers for the multi-agent systems, future expanding the application scope of related research.

Nevertheless, it needs to be emphasized that the aforementioned works did not take input saturation into account, a common constraint in practical systems. Input saturation refers to the limitation imposed by the physical constraints of actuators, when control inputs cannot exceed a certain threshold. This limitation may significantly degrade control performance or even cause instability in the system. Therefore, effectively addressing input saturation has become an important research direction in control theory. To tackle the issue of input saturation, scholars have proposed various solution. For instance, references^{35,36} successfully mitigated the adverse effects of input saturation by introducing smooth function to approximate the saturation and constructing a linear auxiliary system of the same order as the discussed system. References^{37,38} have respectively considered event-triggered adaptive neural network tracking control and convex optimization-based adaptive fuzzy control for nonlinear systems with input saturation. However, these results didn't address fixed-time control. Specifically, the fixed-time control problem for stochastic nonlinear systems with input saturation has not been thoroughly investigated.

Motivated by the above analysis, the presented work focuses on addressing the fixed-time control problem for a kind of nonlinear stochastic systems with input saturation and unknown nonlinearities. The system nonlinearities are assumed unknown and will be handled by FLS approximation method. That is, adding a power integrator technique is used along with FLSs to design the adaptive controller. To deal with the input saturation, the smooth function erf is applied to approximate the saturation function. Then, an adaptive fuzzy fixed-time controller is designed to achieve that all the variables of the considered systems are bounded in probability without violating the input saturation. Based on the proposed improved fixed-time stability criterion, the practical fixed-time stable is analyzed.

The major contributions of this essay could be summarized as below:

- (1) An improved practical fixed-time stability criterion is derived. Based on the practical fixed-time stability criterion for nonlinear system established in reference,²⁹ we have proposed a practical fixed-time stability criterion for stochastic nonlinear systems. Notably, in the proposed criterion the upper estimation of settling time and the residual subset only depend on the design parameters, while the criterion in reference²⁹ has introduced unknown parameters to estimate the upper bound of settling time and the residual subset. Moreover, a more precise upper estimation is given by utilizing the properties of the Gamma function.
- (2) An adaptive fuzzy fixed-time control scheme is developed. Comparing to the finite-time control strategies in references^{23,24,30,35,36} and fixed-time control schemes in references,^{9–11} the piratical fixed-time convergence is considered in this paper for stochastic system simultaneously with the factors of unknown nonlinearities and input saturation. The Gaussian error function and FLSs are respectively applied to approximate input saturation and unknown functions. The adding a power integrator technique is used to develop the control scheme which ensures that all variables are bounded in probability within a fixed time without violating input saturation restriction.

Preliminaries

System description

The following type of stochastic system are considered.

$$\begin{aligned} dx_i &= (x_{i+1} + f_i(\bar{x}_i))dt + g_i^T(\bar{x}_i)d\omega, \\ i &= 1, \dots, n-1, \\ dx_n &= (u(v) + f_n(x))dt + g_n^T(x)d\omega, \end{aligned} \quad (1)$$

where $\bar{x}_i = (x_1, x_2, \dots, x_i)^T \in R^i$, $x = (x_1, \dots, x_n)^T \in R^n$ are denoted as the state variables; ω represents a q dimensional standard Wiener process; $f_i(\bar{x}_i) : R^i \rightarrow R$ and

$g_i(\bar{x}_i) : R^i \rightarrow R^q$ respectively represent system drift and diffusion terms, which are unknown smooth functions with satisfying $f_i(\mathbf{0}) = 0, g_i(\mathbf{0}) = \mathbf{0}$; v indicates control input, and $u(v) \in R$ is called the system control input with saturation.

Notably, $u(v)$ is a saturation control input which is restricted by a maximum amplitude and usually expressed as

$$u(v) = \text{sat}(v) = \begin{cases} -u_M, & v \leq -u_M, \\ v, & -u_M < v < u_M, \\ u_M, & v \geq u_M, \end{cases} \quad (2)$$

where u_M is the unknown upper bound which the value of $|u|$ are required to not exceed.

The goal of this paper is to develop a fixed-time control strategy for system (1) and prove that the designed controller can assure convergence of a small region on system variables while not violating the input saturation restriction.

Saturation transformation

By the definition of $u(v)$, one can find that a sharp corner will appear when $|v| = u_M$. Hence, control input signal $u(v)$ isn't able to be designed directly by a backstepping technique. To tackle this issue, reference⁴ proposed a method, which is employing a smooth function to approximate the saturation function. Thus, we can approximate the saturation function (2) by a smooth function $\mathfrak{h}(v)$ expressed as

$$\mathfrak{h}(v) = u_M * \text{erf}\left(\frac{\sqrt{\pi}v}{2u_M}\right) = u_M * \frac{2}{\sqrt{\pi}} \int_0^{\frac{\sqrt{\pi}v}{2u_M}} e^{-t^2} dt, \quad (3)$$

where $\text{erf}\left(\frac{\sqrt{\pi}v}{2u_M}\right) = u_M * \frac{2}{\sqrt{\pi}} \int_0^{\frac{\sqrt{\pi}v}{2u_M}} e^{-t^2} dt$ is called Gaussian error function.

Therefore, $\text{sat}(v)$ is able to be rewritten as below

$$\text{sat}(v) = \mathfrak{h}(v) + d(v), \quad (4)$$

where $d(v) = \text{sat}(v) - \mathfrak{h}(v)$ represents the approximate error, and the upper bound of which is

$$|d(v)| = |\text{sat}(v) - \mathfrak{h}(v)| < u_M \left| 1 - \frac{2}{\sqrt{\pi}} \int_0^{\frac{\sqrt{\pi}}{2}} e^{-t^2} dt \right| = \aleph. \quad (5)$$

In accordance with the mean-value theorem, one can find a constant μ with $0 < \mu < 1$, such that

$$\mathfrak{h}(v) = \mathfrak{h}(v_0) + \mathfrak{h}'(v_\mu)(v - v_0), \quad (6)$$

where $\mathfrak{h}'(v_\mu) = \frac{\partial \mathfrak{h}(v)}{\partial v} \Big|_{v=v_\mu} = e^{-\left(\frac{\sqrt{\pi}v}{2u_M}\right)^2} \Big|_{v=v_\mu}$, $v_\mu = \mu v + (1 - \mu)v_0$. By selecting $v_0 = 0$, one gains $\mathfrak{h}(v_0) = \mathfrak{h}(0) = 0$ and redescibes equation (6) as

$$\mathfrak{h}(v) = \mathfrak{h}'(v_\mu)v. \quad (7)$$

Hence, equation (4) is reexpressed as

$$\text{sat}(v) = \mathfrak{h}'(v_\mu)v + d(v). \quad (8)$$

Remark 1. It should be pointed out that the smooth function applied in this paper is erf , while the $\tanh$ function is adopted in other result. It is easy to find that the erf has a better approximation effect than the $\tanh$ function, which can be observed from Figure 1. Although the $\tanh$ function and erf function have smooth transition characteristics when the input is close to saturation, the curvature of $\tanh$ function transition region is larger, which may lead to larger response errors. Compared with $\tanh$ function, erf function can better approximate the saturation boundary.

Assumption 1. There is a positive constant γ that makes

$$\gamma \leq \mathfrak{h}'(v_\mu) \leq 1. \quad (9)$$

Remark 2. This assumption is reasonable. It's easily known from equation (6) that $\mathfrak{h}'(v_\mu) = \frac{\partial \mathfrak{h}(v)}{\partial v} \Big|_{v=v_\mu} = e^{-\left(\frac{\sqrt{\pi}v}{2u_M}\right)^2} \Big|_{v=v_\mu}$, $v_\mu = \mu v + (1 - \mu)v_0$. By selecting $v_0 = 0$, one gains $\mathfrak{h}'(v_\mu) = e^{-\left(\frac{\sqrt{\pi}v}{2u_M}\right)^2}$, where $\mu \in (0, 1)$ is a constant and $|v| < u_M$. Clearly, $\mathfrak{h}'(v_\mu)$ increases with $v \in (-u_M, 0)$ and decrease with $v \in (0, u_M)$. It follows that $e^{-\left(\frac{\sqrt{\pi}\mu}{2}\right)^2} < \mathfrak{h}'(v_\mu) \leq 1$. Due to $\mu \in (0, 1)$, we can calculate that $\mathfrak{h}'(v_\mu) > e^{-\left(\frac{\sqrt{\pi}}{2}\right)^2}$. Therefore, we can take the value of γ as $\gamma \leq e^{-\left(\frac{\sqrt{\pi}}{2}\right)^2}$.

Definitions and Lemmas

We introduce some relevant Definitions and Lemmas, which provide a theoretical basis for subsequent controller design and stability analysis.

Consider stochastic systems as follows:

$$dx = f(x)dt + g(x)d\omega, \forall x(0) = x_0 \in R^n, \quad (10)$$

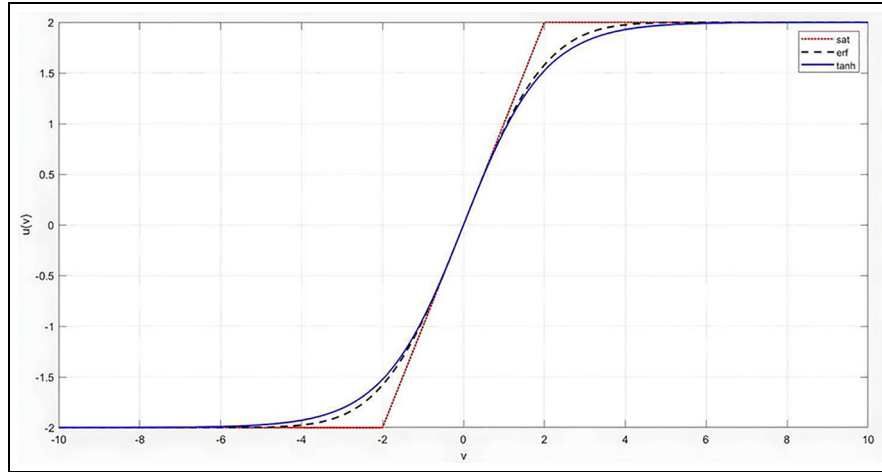


Figure 1. Two approximation curves of saturation function.

where $x \in R^n$ represents the state vector, ω denotes a q dimensional standard Wiener process; $f(x): R^n \rightarrow R$ and $g(x): R^n \rightarrow R^q$ are continuous functions meeting $f(0) = 0, g(0) = 0$.

Definition 1.³⁹ For $\forall V(x) \in C^2$ with respect to system (10), the differential operator of $V(x)$ is expressed As

$$\mathcal{L}V = \frac{\partial V}{\partial x}f(x) + \frac{1}{2}tr\left\{g^T(x)\frac{\partial^2 V}{\partial x^2}g(x)\right\}, \quad (11)$$

where $\frac{1}{2}tr\left\{g^T(x)\frac{\partial^2 V}{\partial x^2}g(x)\right\}$ is the Hessian term.

Lemma 1.²² For system (10), if there exists a C^2 function $V(x): R^n \rightarrow R^+$, for $\forall \mu_1, \mu_2 \in R^+, 0 < \sigma < 1, \beta > 1$, such that

$$\mathcal{L}V \leq -\mu_1 V^\sigma(x) - \mu_2 V^\beta(x), \forall x \in R^n \setminus \{0\}. \quad (12)$$

Then, the origin point $x = 0$ of system (10) is fixed-time stable in probability with the settling time T_{x_0} fulfilling

$$E(T_{x_0}) \leq \frac{\Gamma\left(\frac{1-\sigma}{\beta-\sigma}\right)\Gamma\left(\frac{\beta-1}{\beta-\sigma}\right)\left(\frac{\mu_1}{\mu_2}\right)^{\frac{1-\sigma}{\beta-\sigma}}}{\mu_1(\beta-\sigma)}, \forall x_0 \in R^n \setminus \{0\}. \quad (13)$$

Lemma 2.⁴⁰ For $\chi_1 \in R$ and constant $\check{\delta}_1 > 0$, there is

$$0 \leq |\chi_1| < \check{\delta}_1 + \frac{\chi_1^2}{\sqrt{\chi_1^2 + \check{\delta}_1^2}}. \quad (14)$$

Lemma 3.³⁹ For $\forall(\chi_1, \chi_2) \in R^2$, it holds that

$$\chi_1 \chi_2 \leq \frac{\check{\delta}_1^{p_1}}{p_1} |\chi_1|^{p_1} + \frac{1}{q_1 \check{\delta}_1^{q_1}} |\chi_2|^{q_1}. \quad (15)$$

where $\check{\delta}_1 > 0, p_1 > 1, q_1 > 1$ are constants and $\frac{1}{p_1} + \frac{1}{q_1} = 1$.

Lemma 4.²² For every $q_1 \in R^+$, and variables $\chi_i \in R, i = 1, \dots, n$, we have

$$\left(\sum_{i=1}^n |\chi_i|\right)^{q_1} \leq \max\{1, n^{q_1-1}\} \left(\sum_{i=1}^n |\chi_i|^{q_1}\right). \quad (16)$$

Lemma 5.⁷ For $\forall \chi_1, \chi_2 \in R, 0 < q_1 < 1$ satisfying $\chi_2 > 2\chi_1$, one has

$$\chi_2^{q_1} - \chi_1^{q_1} \geq (2^{q_1} - 1)(\chi_2 - \chi_1)^{q_1}. \quad (17)$$

Lemma 6.⁶ Let $\chi_1, \chi_2 \in R$. For $\forall$ positive real numbers q_4, q_5 , and any functions $\mathcal{C}(\cdot), \mathcal{Q}(\cdot) \in R^+$, one has

$$\begin{aligned} \mathcal{C}(\cdot)|\chi_1|^{q_4}|\chi_2|^{q_5} &\leq \frac{q_4 \mathcal{Q}(\cdot)}{q_4 + q_5} |\chi_1|^{q_4 + q_5} \\ &+ \frac{q_5 (\mathcal{C}(\cdot))^{\frac{q_4 + q_5}{q_5}} (\mathcal{Q}(\cdot))^{\frac{q_4}{q_5}}}{q_4 + q_5} |\chi_2|^{q_4 + q_5}. \end{aligned} \quad (18)$$

Lemma 7.²⁴ Let $\mathcal{G}(x)$ be a continuous function defined on a compact set Λ , then for an arbitrary small constant $\tau > 0$, there exists a FLS $\mathcal{Y}(x) = \mathcal{H}^T \Psi(x)$ satisfying

$$\sup_{x \in \Lambda} |\mathcal{G}(x) - \mathcal{H}^T \Psi(x)| \leq \tau, \quad (19)$$

where $\Psi(x) = (\psi_1(x), \dots, \psi_N(x))^T$ is the fuzzy basis function, $\mathcal{H} = (o_1, \dots, o_N)^T$ is the optimal parameter vector, and N is the fuzzy rules number.

Remark 3. From Lemma 7 we know that every continuous function $\mathcal{G}(x)$ defined on a compact set Λ is capable of being formulated as

$$\mathcal{G}(x) = \mathcal{H}^T \Psi(x) + \delta(x), \quad (20)$$

where $\delta(x)$ is approximation error and satisfies $|\delta(x)| < \tau$.

Main results

Based on the control goal of the considered system, we know that means the considered system is practical fixed-time stable. Thus, we need to use the theorem of practical fixed-time stability criterion. In this part, we firstly give a theorem of an improved practical fixed-time stability criterion. Then, a fixed-time control strategy is derived. Subsequently, a feedback control is designed by using the adding a power integral technique and the approximated method of FLS. Finally, based on the given stability criterion, the fixed-time convergence of the considered system is rigorously analyzed.

Improved practical fixed-time stability criterion

Theorem 1. For system (10), suppose that there exists a C^2 function $V(x): R^n \rightarrow R^+$ such that

$$\mathcal{L}V(x) \leq -\mu_1 V^\sigma(x) - \mu_2 V^\beta(x) + \rho, \quad (21)$$

with $\sigma \in (0, 1), \beta > 1; \mu_1, \mu_2, \rho$ being positive parameters, then system (10) is practical fixed-time stable in probability. The corresponding settling time $T_p(x_0)$ meets

$$E(T_p(x_0)) \leq T_{\max} := \frac{\Gamma(\frac{1-\sigma}{\beta-\sigma})\Gamma(\frac{\beta-1}{\beta-\sigma})}{(2^\sigma-1)\mu_1(\beta-\sigma)} \left(\frac{(2^\sigma-1)\mu_1}{\mu_2} \right)^{\frac{1-\sigma}{\beta-\sigma}}.$$

Proof: Borrowing from the proving method in reference,¹² we prove this theorem in two situations. Firstly, let $Y(\chi) = \mu_1 \chi^\sigma + \mu_2 \chi^\beta$, then we know that the range of $Y(\chi)$ is $(0, +\infty)$. Thus, for any $\rho > 0$, there must exist a constant $\hbar > 0$ such that $\rho = \mu_1 \hbar^\sigma + \mu_2 \hbar^\beta$. Consequently, equation (21) can be reexpressed as

$$\mathcal{L}V(x) \leq -\mu_1 (V^\sigma(x) - \hbar^\sigma) - \mu_2 (V^\beta(x) - \hbar^\beta). \quad (22)$$

Now, we divide the domain of the state variable into two regions: $\Xi_1 = \{x | V(x) \geq 2\hbar\}$ and $\Xi_2 = \{x | V(x) < 2\hbar\}$.

Case 1: When $x \in \Xi_1$, let $\mathfrak{S} = V - \hbar$. Obviously, $\mathfrak{S}$ is positive definite on Ξ_1 . Utilizing Lemma 5 yields

$$\begin{aligned} \mathcal{L}\mathfrak{S}(x) &= \mathcal{L}V(x) - \mathcal{L}\hbar \\ &\leq -\mu_1 (V^\sigma(x) - \hbar^\sigma) - \mu_2 (V^\beta(x) - \hbar^\beta) \\ &\leq -\mu_1 (2^\sigma - 1)(V(x) - \hbar)^\sigma - \mu_2 (V(x) - \hbar)^\beta \\ &= -\mu_1 (2^\sigma - 1)\mathfrak{S}^\sigma(x) - \mu_2 \mathfrak{S}^\beta(x). \end{aligned} \quad (23)$$

In Accordance with Lemma 1, the system trace will reach Ξ_2 before the settling time T_p , which satisfies

$$ET_p \leq T_{\max} := \frac{\Gamma(\frac{1-\sigma}{\beta-\sigma})\Gamma(\frac{\beta-1}{\beta-\sigma})}{(2^\sigma-1)\mu_1(\beta-\sigma)} \left[\frac{(2^\sigma-1)\mu_1}{\mu_2} \right]^{\frac{1-\sigma}{\beta-\sigma}}. \quad (24)$$

Case 2: When $x \in \Xi_2$, the trajectory of the system state variable does not exceed Ξ_2 , so it is easily known that the state variable is bounded. Hence, system (10) is practical fixed-time stable in probability and the system trace converges to $\Xi_2 = \{x | V(x) < 2\hbar\}$. Moreover, if $0 < \hbar < 1$, then it holds that $\hbar < \hbar^\sigma < \frac{\rho}{\mu_1}$, while it holds that $1 < \hbar < \hbar^\beta < \frac{\rho}{\mu_2}$ for $\hbar > 1$. In other words, the system trace converges to $\{x | V(x) \leq \min\{\frac{2\rho}{\mu_1}, \frac{2\rho}{\mu_2}\}\}$.

Remark 4. It's important to note that the practical fixed-time stability criterion is also given for deterministic nonlinear system in reference,²⁹ but the settling time T_2 satisfy $T_2 \leq \tilde{T}_{\max} = \frac{1}{\mu_1(1-\sigma)\eta} + \frac{1}{\mu_2(\beta-1)\eta}$. According to the proof procedures of the two criteria, one easily gains $\tilde{T}_{\max} \geq T_{\max}$. In additional, the residual subset $\left\{x | V(x) \leq \min\left\{\left(\frac{\rho}{\mu_1(1-\eta)}\right)^{\frac{1}{\sigma}}, \left(\frac{\rho}{\mu_2(1-\eta)}\right)^{\frac{1}{\beta}}\right\}\right\}$ obtained in reference²⁹ is also affected by the extra parameter η . Whereas, in this paper, we convert the inequality into a standard form and calculating the upper bound estimation and residual subset from the design parameters without using other parameters.

On the other hand, if one wants to consider the fixed-time control problem of stochastic nonlinear system with some uncertain parameters, we can extend the proposed method to address this issue. However, due to the uncertain parameters, the control procedure should consider introducing another adaptive law. So one can evaluate the value of upper bound for the uncertain systems in a way similar to this article, but the meanings of the design parameters are different.

The design of state-feedback controller

In this sub-section, we will design the appropriate controller by explicit steps. Firstly, the subsequent coordinate transformations are given

$$\begin{cases} z_1 = x_1, \\ z_i = x_i - \zeta_i, 2 \leq i \leq n, \end{cases} \quad (25)$$

where $\zeta_2, \dots, \zeta_n$ are virtual controllers and will be constructed afterwards. Next, for convenience and without causing confusion, we sometimes omit the variables of the functions, for example, $f_i(\bar{x}_i)$ is abbreviated as f_i . Hence, system (1) can be redescribed as

$$\begin{aligned} dz_1 &= (z_2 + f_1 + \zeta_2)dt + g_1^T(x_1)d\omega, \\ dz_i &= (z_{i+1} + f_i - \mathcal{L}\zeta_i + \zeta_{i+1})dt + \mathcal{I}_i^T(Z_i)d\omega, \\ dz_n &= (u(v) + f_n - \mathcal{L}\zeta_n)dt + \mathcal{I}_n^T(Z_n)d\omega, \end{aligned} \quad (26)$$

where $\mathcal{I}_i(Z_i) = g_i - \sum_{j=1}^{i-1} \frac{\partial \zeta_i}{\partial x_j} g_j$, $\mathcal{L}\zeta_i = \sum_{j=1}^{i-1} \frac{\partial \zeta_i}{\partial x_j} (\dot{x}_{j+1} + f_j) + \sum_{j=1}^{i-1} \frac{\partial \zeta_i}{\partial \theta_j} \dot{\theta}_j + \frac{1}{2} \sum_{j,k=1}^{i-1} \frac{\partial^2 \zeta_i}{\partial x_j \partial x_k} g_j^T g_k$, $2 \leq i \leq n-1$, and the definition of $Z_i, \hat{\theta}_j, \dot{\theta}_j$ will be subsequently provided.

Step 1. Select Lyapunov function

$$V_1 = \frac{1}{4}z_1^4 + \frac{1}{2\lambda_1}\tilde{\theta}_1^2, \quad (27)$$

where $\lambda_1 > 0$ is a constant parameter; $\tilde{\theta}_1 = \theta_1 - \hat{\theta}_1$ denotes the estimation error of the unknown parameter θ_1 with $\hat{\theta}_1$ being the estimation of θ_1 , which will be given afterwards.

From equation (11), one can infer

$$\mathcal{L}V_1 = z_1^3(z_2 + \zeta_2 + f_1) + \frac{3}{2}z_1^2g_1^Tg_1 - \frac{1}{\lambda_1}\tilde{\theta}_1\dot{\theta}_1. \quad (28)$$

It follows from Lemma 3 that

$$\begin{aligned} |z_1^3z_2| &\leq \frac{3}{4}z_1^4 + \frac{1}{4}z_2^4, \\ \left| \frac{3}{2}z_1^2g_1^Tg_1 \right| &\leq \frac{3}{4}\nu_1^{-2}z_1^4 \|g_1\|^4 + \frac{3}{4}\nu_1^2, \end{aligned} \quad (29)$$

where $\nu_1 \geq 0$ is a constant parameter.

Then, we let $X_1 = x_1$ and denote

$$\mathcal{G}_1(X_1) = f_1 + \frac{3}{4}\nu_1^{-2}z_1 \|g_1(x_1)\|^4,$$

From Lemma 7, we know that $\mathcal{G}_1(X_1)$ can be approximated by FLS $\mathcal{H}_1^T\psi_1(X_1)$, that is,

$$\mathcal{G}_1(X_1) = \mathcal{H}_1^T\psi_1(X_1) + \delta_1(X_1), \quad (30)$$

where $|\delta_1(X_1)| \leq \tau_1$ is approximation error with $\tau_1 > 0$ being a constant.

Letting $\mathcal{H}_1^T\mathcal{H}_1 = \theta_1$ and using lemma 3, one can gain

$$\begin{aligned} |z_1^3\mathcal{G}_1(X_1)| &= |z_1^3(\mathcal{H}_1^T\psi_1(X_1) + \delta_1(X_1))| \\ &\leq \frac{1}{2a_1^2}z_1^6\theta_1 \|\psi_1(X_1)\|^2 + \frac{a_1^2}{2} + \frac{3}{4}z_1^4 + \frac{\tau_1^4}{4}, \end{aligned} \quad (31)$$

where $a_1 \geq 0$ is a design parameter.

Applying equation (31) in (28) leads to

$$\begin{aligned} \mathcal{L}V_1 &\leq z_1^3\left(\zeta_2 + \frac{1}{2a_1^2}z_1^3\theta_1 \|\psi_1(X_1)\|^2\right) + \frac{a_1^2}{2} + \frac{3}{2}z_1^4 \\ &\quad + \frac{1}{4}z_2^4 + \frac{\tau_1^4}{4} - \frac{1}{\lambda_1}\tilde{\theta}_1\dot{\theta}_1 + \frac{3}{4}\nu_1^2. \end{aligned} \quad (32)$$

So one can design the adaptive law and virtual controller as

$$\begin{aligned} \dot{\hat{\theta}}_1 &= \frac{\lambda_1}{2a_1^2}z_1^6 \|\psi_1\|^2 - b_1\hat{\theta}_1, \quad \zeta_2 = -\frac{z_1^3\bar{\xi}_1^2}{\sqrt{z_1^6\bar{\xi}_1^2 + \varrho_1^2}} \\ &\quad - \frac{k_2z_1^{4\beta-3}}{4\beta} - \frac{z_1^3\hat{\theta}_1\psi_1^T\psi_1}{2a_1^2} - \frac{3z_1}{2}, \end{aligned} \quad (33)$$

where $\bar{\xi}_1 = \frac{k_1}{4^\sigma}z_1^{4\sigma-3}$; and $k_1 > 0, k_2 > 0, b_1 > 0, \varrho_1 > 0$ are all constant parameters.

Besides, it can be concluded from Lemma 2 that

$$z_1^3\left(-\frac{z_1^3\bar{\xi}_1^2}{\sqrt{z_1^6\bar{\xi}_1^2 + \varrho_1^2}}\right) = -\frac{z_1^6\bar{\xi}_1^2}{\sqrt{z_1^6\bar{\xi}_1^2 + \varrho_1^2}} \leq \varrho_1 - z_1^3\bar{\xi}_1. \quad (34)$$

Thus, substituting (33) and (34) into (32) yields

$$\begin{aligned} \mathcal{L}V_1 &\leq -\frac{k_1}{4^\sigma}z_1^{4\sigma} - \frac{k_2}{4\beta}z_1^{4\beta} - \frac{b_1}{\lambda_1}\tilde{\theta}_1\hat{\theta}_1 + \frac{1}{4}z_2^4 \\ &\quad + \frac{3}{4}\nu_1^2 + \frac{a_1^2}{2} + \frac{\tau_1^4}{4} + \varrho_1. \end{aligned} \quad (35)$$

Furthermore, it's not difficult to obtain

$$\frac{b_1}{\lambda_1}\tilde{\theta}_1\hat{\theta}_1 = \frac{b_1}{\lambda_1}\tilde{\theta}_1(\theta_1 - \hat{\theta}_1) \leq -\frac{b_1}{2\lambda_1}\tilde{\theta}_1^2 + \frac{b_1}{2\lambda_1}\theta_1^2. \quad (36)$$

Then, taking (36) into (35), one can gain that

$$\mathcal{L}V_1 \leq -\frac{k_1}{4^\sigma}z_1^{4\sigma} - \frac{k_2}{4\beta}z_1^{4\beta} - \frac{b_1}{2\lambda_1}\tilde{\theta}_1^2 + \Delta_1 + \frac{1}{4}z_2^4, \quad (37)$$

where $\Delta_1 = \frac{3}{4}\nu_1^2 + \frac{a_1^2}{2} + \frac{\tau_1^4}{4} + \frac{b_1}{2\lambda_1}\theta_1^2 + \varrho_1$ is a positive constant.

Step $i(2 \leq i \leq n-1)$. We can assume that there is a Lyapunov function $V_{i-1} = \sum_{j=1}^{i-1} \frac{1}{4}z_j^4 + \sum_{j=1}^{i-1} \frac{1}{2\lambda_j}\tilde{\theta}_j^2$ in step $i-1$, along with a string of virtual controllers and adaptive laws $(\zeta_{j+1}, \dot{\hat{\theta}}_j)(j=1, \dots, i-1)$, which can make

$$\begin{aligned} \mathcal{L}V_{i-1} \leq & -\frac{k_1}{4^\sigma} \sum_{j=1}^{i-1} z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^{i-1} z_j^{4\beta} - \sum_{j=1}^{i-1} \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 \\ & + \sum_{j=1}^{i-1} \Delta_j + \frac{z_i^4}{4}, \end{aligned}$$

with b_j, λ_j being positive constant parameters.

Thus, we can construct next indicated Lyapunov function

$$V_i = V_{i-1} + \frac{1}{4}z_i^4 + \frac{1}{2\lambda_i}\tilde{\theta}_i^2, \quad (38)$$

where the parameter λ_i is a positive constant; $\tilde{\theta}_i = \theta_i - \hat{\theta}_i$ represents the estimation error of the unknown parameter θ_i with $\hat{\theta}_i$ being the estimation of θ_i , the definition of which will be given afterwards.

Then, it can be concluded that

$$\begin{aligned} \mathcal{L}V_i \leq & \mathcal{L}V_{i-1} + z_i^3(z_{i+1} + \zeta_{i+1} + f_i - \mathcal{L}\zeta_i) \\ & - \frac{\tilde{\theta}_i \dot{\hat{\theta}}_i}{\lambda_i} + \frac{3z_i^2}{2}\mathcal{I}_i^T \mathcal{I}_i. \end{aligned} \quad (39)$$

According to Lemma 3, it can be inferred that

$$|z_i^3 z_{i+1}| \leq \frac{3}{4}z_i^4 + \frac{1}{4}z_{i+1}^4, \quad (40)$$

$$\left| \frac{3z_i^2}{2}\mathcal{I}_i^T \mathcal{I}_i \right| \leq \frac{3z_i^4}{4\nu_i^2} \|\mathcal{I}_i\|^4 + \frac{3\nu_i^2}{4}. \quad (41)$$

Hence, taking (40), (41) into (39), one can attain that

$$\begin{aligned} \mathcal{L}V_i \leq & -\frac{k_1}{4^\sigma} \sum_{j=1}^{i-1} z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^{i-1} z_j^{4\beta} - \sum_{j=1}^{i-1} \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 \\ & + z_i^3 \left(\zeta_{i+1} + f_i - \mathcal{L}\zeta_i + \frac{3z_i}{4\nu_i^2} \|\mathcal{I}_i\|^4 \right) \\ & + \sum_{j=1}^{i-1} \Delta_j + \frac{1}{4}z_{i+1}^4 + \frac{3}{4}\nu_i^2 - \frac{1}{\lambda_i} \tilde{\theta}_i \dot{\hat{\theta}}_i + z_i^4. \end{aligned} \quad (42)$$

Thereby, it can be recorded as

$$\mathcal{G}_i(Z_i) = f_i - \mathcal{L}\zeta_i + \frac{3}{4}\nu_i^{-2}z_i \|\mathcal{I}_i\|^4,$$

where $X_i = \left(\bar{x}_i^T, \bar{\theta}_i^T \right)^T$, $\bar{\theta}_i = (\hat{\theta}_1, \dots, \hat{\theta}_i)^T$. Similarly, $\mathcal{G}_i(X_i)$ can be approximated by FLS $\mathcal{H}_i^T \psi_i(X_i)$, that is,

$$\mathcal{G}_i(X_i) = \|\mathcal{H}_i^T \psi_i(X_i)\| + \delta_i(X_i), \quad (43)$$

where $\delta_i(X_i) \leq \tau_i$ is approximation error with constant $\tau_i > 0$.

Letting $\theta_i = \mathcal{H}_i^T \mathcal{H}_i$, we can obtain from Lemma 3 that

$$\begin{aligned} |z_i^3 \mathcal{G}_i(X_i)| &= |z_i^3 (\mathcal{H}_i^T \psi_i(X_i) + \delta_i(X_i))| \\ &\leq \frac{1}{2a_i^2} z_i^6 \theta_i \|\psi_i(X_i)\|^2 + \frac{a_i^2}{2} + \frac{3}{4}z_i^4 + \frac{\tau_i^4}{4}, \end{aligned} \quad (44)$$

where $a_i > 0$ denotes a design constant parameter.

By using equation (44), one can infer

$$\begin{aligned} \mathcal{L}V_i \leq & -\frac{k_1}{4^\sigma} \sum_{j=1}^{i-1} z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^{i-1} z_j^{4\beta} - \sum_{j=1}^{i-1} \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 \\ & + z_i^3 \left(\zeta_i + \frac{1}{2a_i^2} z_i^3 \theta_i \|\psi_i(X_i)\|^2 \right) + \frac{7}{4}z_i^4 + \frac{1}{4}z_{i+1}^4 \\ & + \frac{a_i^2}{2} + \frac{\tau_i^4}{4} + \frac{3}{4}\nu_i^2 - \frac{1}{\lambda_i} \tilde{\theta}_i \dot{\hat{\theta}}_i + \sum_{j=1}^{i-1} \Delta_j. \end{aligned} \quad (45)$$

Thus, we design the adaptive law and virtual controller as

$$\begin{aligned} \dot{\hat{\theta}}_i &= \frac{\lambda_i}{2a_i^2} z_i^6 \|\psi_i(X_i)\|^2 - b_i \hat{\theta}_i, \\ \zeta_{i+1} &= -\frac{z_i^3 \bar{\xi}_i^2}{\sqrt{z_i^6 \bar{\xi}_i^2 + \varrho_i^2}} - \frac{k_2}{4^\beta} z_i^{4\beta-3} - \frac{z_i^3 \hat{\theta}_i \|\psi_i(X_i)\|^2}{2a_i^2} - \frac{7z_i}{4}, \end{aligned} \quad (46)$$

where $\bar{\xi}_i = \frac{k_1}{4^\sigma} z_i^{4\sigma-3}$, b_i and ϱ_i are both positive parameters.

Consequently, it can be deduced from Lemma 2 that

$$z_i^3 \left(-\frac{z_i^3 \bar{\xi}_i^2}{\sqrt{z_i^6 \bar{\xi}_i^2 + \varrho_i^2}} \right) = -\frac{z_i^6 \bar{\xi}_i^2}{\sqrt{z_i^6 \bar{\xi}_i^2 + \varrho_i^2}} \leq \varrho_i - z_i^3 \bar{\xi}_i. \quad (47)$$

Hence, taking (46) and (47) into (45), we can achieve that

$$\begin{aligned}
\mathcal{L}V_i &\leq -\frac{k_1}{4^\sigma} \sum_{j=1}^{i-1} z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^{i-1} z_j^{4\beta} - \sum_{j=1}^{i-1} \frac{b_j}{2\lambda_j} \tilde{\theta}_i^2 - \frac{k_1}{4^\sigma} z_i^{4\sigma} \\
&\quad - \frac{k_2}{4^\beta} z_i^{4\beta} + \frac{1}{2a_i^2} z_i^6 (\theta_i - \hat{\theta}_i - \tilde{\theta}) \psi_i^T \psi_i \\
&\quad + \sum_{j=1}^{i-1} \Delta_j + \frac{1}{4} z_{i+1}^4 + q_i + \frac{a_i^2}{2} + \frac{\tau_i^4}{4} + \frac{3}{4} \nu_i^2 + \frac{b_i}{\lambda_i} \tilde{\theta}_i \hat{\theta}_i \\
&\leq -\frac{k_1}{4^\sigma} \sum_{j=1}^i z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^i z_j^{4\beta} - \sum_{j=1}^{i-1} \frac{b_j}{2\lambda_j} \tilde{\theta}_i^2 + \sum_{j=1}^{i-1} \Delta_j \\
&\quad + \frac{1}{4} z_{i+1}^4 + q_i + \frac{a_i^2}{2} + \frac{\tau_i^4}{4} + \frac{3}{4} \nu_i^2 + \frac{b_i}{\lambda_i} \tilde{\theta}_i \hat{\theta}_i.
\end{aligned} \tag{48}$$

It's not difficult to discover

$$\frac{b_i}{\lambda_i} \tilde{\theta}_i \hat{\theta}_i = \frac{b_i}{\lambda_i} \tilde{\theta}_i (\theta_i - \tilde{\theta}_i) \leq \frac{b_i}{2\lambda_i} \theta_i^2 - \frac{b_i}{2\lambda_i} \tilde{\theta}_i^2. \tag{49}$$

Substituting (49) into (48) yields

$$\begin{aligned}
\mathcal{L}V_i &\leq -\frac{k_1}{4^\sigma} \sum_{j=1}^i z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^i z_j^{4\beta} - \sum_{j=1}^i \frac{b_j}{2\lambda_j} \tilde{\theta}_i^2 \\
&\quad + \sum_{j=1}^i \Delta_j + \frac{1}{4} z_{i+1}^4,
\end{aligned} \tag{50}$$

where $\Delta_i = \frac{3}{4} \nu_i^2 + \frac{a_i^2}{2} + \frac{\tau_i^4}{4} + \frac{b_i}{2\lambda_i} \theta_i^2 + q_i$ is a positive constant.

Last Step. In light of above induction step, we are capable of attaining a string of virtual controllers and adaptive laws, so the final Lyapunov function V_n should be established as

$$V_n = V_{n-1} + \frac{1}{4} z_n^4 + \frac{1}{2\lambda_n} \tilde{\theta}_n^2, \tag{51}$$

where $\lambda_n \geq 0$ is a constant parameter, and $\tilde{\theta}_n = \theta_n - \hat{\theta}_n$ is the estimation error of the unknown parameter θ_n with $\hat{\theta}_n$ being the estimation of θ_n , the definition of which will be given afterwards.

From Definition 1, equations (8) and (26), we can get

$$\begin{aligned}
\mathcal{L}V_n &\leq \mathcal{L}V_{n-1} + z_n^3 \left(\mathfrak{h}'(\nu_\mu) + d(v) + f_n - \mathcal{L}\zeta_n \right) \\
&\quad - \frac{\tilde{\theta}_n \dot{\theta}_n}{\lambda_n} + \frac{3z_n^2}{2} \mathcal{I}_n^T \mathcal{I}_n.
\end{aligned} \tag{52}$$

Then, according to (5) and Lemma 3, one gains

$$|z_n^3 d(v)| \leq |z_n^3 \aleph| \leq \frac{3}{4} z_n^4 + \frac{1}{4} \aleph^4, \tag{53}$$

Therefore, substituting equations (48), (53) and (54) into (52) yields

$$\begin{aligned}
\mathcal{L}V_n &\leq -\frac{k_1}{4^\sigma} \sum_{j=1}^{n-1} z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^{n-1} z_j^{4\beta} - \sum_{j=1}^{n-1} \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 \\
&\quad + z_n^3 \left(f_n - \mathcal{L}\zeta_n + \mathfrak{h}'(\nu_\mu) v + \frac{3z_n}{4\nu_n^2} \|\mathcal{I}_n\|^4 \right) \\
&\quad + \frac{3}{4} \nu_n^2 - \frac{1}{\lambda_n} \tilde{\theta}_n \dot{\theta}_n + \frac{7}{4} z_n^4 + \frac{1}{4} \aleph^4 + \sum_{j=1}^{n-1} \Delta_j.
\end{aligned} \tag{55}$$

Then, $\mathcal{G}_n(X_n)$ is denoted as

$$\mathcal{G}_n(X_n) = f_n - \mathcal{L}\zeta_n + \frac{3}{4} \frac{z_n}{\nu_n^2} \|\mathcal{I}_n\|^4,$$

where $X_n = \left(\bar{x}_n^T, \bar{\theta}_n^T \right)^T$, $\bar{\theta}_n = (\hat{\theta}_1, \dots, \hat{\theta}_n)^T$. Obviously, $\mathcal{G}(X_n)$ can be approximated by using FLS $\mathcal{H}_n^T \psi_n$, that is,

$$\mathcal{G}_n(X_n) = \|\mathcal{H}_n^T \psi_n(X_n)\| + \delta_n(X_n), \tag{56}$$

where $\delta_n(Z_n) \leq \tau_n$ is approximation error with a constant $\tau_n > 0$.

Let $\theta_n = \mathcal{H}_n^T \mathcal{H}_n$, and then we can infer from Lemma 3 that

$$\begin{aligned}
|z_n^3 \mathcal{G}_n(X_n)| &= |z_n^3 (\mathcal{H}_n^T \psi_n(X_n) + \delta_n(X_n))| \\
&\leq \frac{z_n^6 \theta_n}{2a_n^2} \|\psi_n(X_n)\|^2 + \frac{a_n^2}{2} + \frac{3}{4} z_n^4 + \frac{\tau_n^4}{4},
\end{aligned} \tag{57}$$

where $a_n > 0$ is constant design parameter.

Thus, taking (57) into (55), one can obtain that

$$\begin{aligned}
\mathcal{L}V_n &\leq -\frac{k_1}{4^\sigma} \sum_{j=1}^{n-1} z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^{n-1} z_j^{4\beta} - \sum_{j=1}^{n-1} \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 \\
&\quad + z_n^3 \left(\mathfrak{h}'(\nu_\mu) v + \frac{1}{2a_n^2} z_n^3 \theta_n \|\psi_n(Z_n)\|^2 \right) \\
&\quad + \frac{a_n^2}{2} + \sum_{j=1}^{n-1} \Delta_j + \frac{\tau_n^4}{4} + \frac{3}{4} \nu_n^2 - \frac{1}{\lambda_n} \tilde{\theta}_n \dot{\theta}_n + \frac{7}{4} z_n^4 + \frac{1}{4} \aleph^4.
\end{aligned} \tag{58}$$

Accordingly, we are able to construct the last adaptive law and the actual controller as

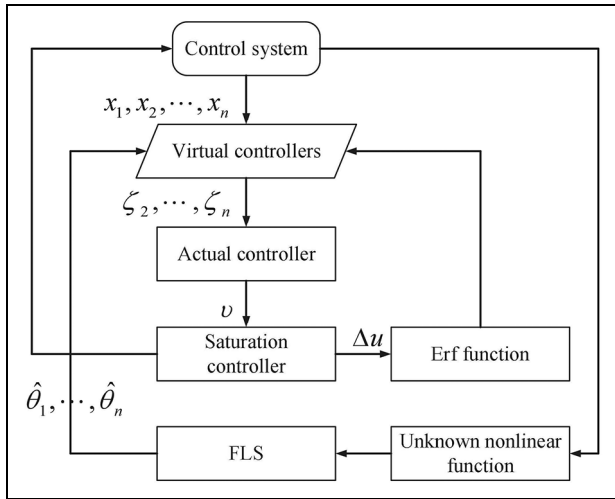


Figure 2. The block diagram of the proposed scheme.

$$\begin{aligned}\dot{\theta}_n &= \frac{\lambda_1}{2a_n^2} z_n^6 \|\psi_n(X_n)\|^2 - b_n \hat{\theta}_n, \\ v &= \zeta_{n+1} \\ &= -\frac{1}{\gamma} \left(\frac{z_n^3 \bar{\xi}_n^2}{\sqrt{z_n^6 \bar{\xi}_n^2 + \varrho_n^2}} + \frac{k_2}{4\beta} z_n^{4\beta-3} + \frac{1}{2a_n^2} z_n^3 \hat{\theta}_n \|\psi_n(X_n)\|^2 + \frac{7}{4} z_n \right),\end{aligned}\quad (59)$$

where $\bar{\xi}_n = \frac{k_1}{4^\sigma} z_n^{4\sigma-3}$; $b_n, \varrho_n \geq 0$ are constant parameters.

Till up, the controller design is finished. And we give a block diagram (Figure 2) of the proposed method to illustrate the control design process.

Remark 5. It should be mentioned that the system nonlinearities are unknown. To tackle this problem, every unknown function $\mathcal{G}_1(\cdot)$ appearing in the design procedure is approximated by the fuzzy logic system $\mathcal{H}^T \Psi(\cdot)$, which arises the unknown parameter θ_i with the estimation $\hat{\theta}_i$. By designing series of adaptive laws of $\hat{\theta}_i$'s, the actual control is correspondingly constructed.

Stability analysis

Theorem 2. Suppose system (1) fulfills Assumption 1, then, under the proposed controller and adaptive law (59) along with a string of virtual controllers and adaptive laws $(\zeta_{i+1}, \hat{\theta}_i)$ ($i = 1, \dots, n-1$), all the signals of considered system are practical fixed-time stable in probability. Meanwhile, the related stochastic settling time T_p meets $ET_p \leq T_{\max} := \frac{\Gamma(\frac{1-\sigma}{\beta-\sigma})\Gamma(\frac{\beta-1}{\beta-\sigma})}{(2^\sigma-1)\mu_1(\beta-\sigma)} \left[\frac{(2^\sigma-1)\mu_1}{\mu_2} \right]^{\frac{1}{\beta-\sigma}}$, where $\mu_1 = \min\{k_1, \frac{b^\sigma}{2}\}$, $\mu_2 = n^{1-\beta}\bar{\mu}_2$, $\bar{\mu}_2 = \min\{n^{1-\beta}k_2, 2^{-\beta}n^{1-\beta}b\}$, $b = \min_{1 \leq j \leq n} \{b_j\}$

Proof: Firstly, Let $Z(t) = (z_1, \dots, z_n, \hat{\theta}_1, \dots, \hat{\theta}_n)^T$, then we easily have $V_n(X_n(t)) = V_n(Z(t))$.

Using Lemma 2, we can get

$$z_n^3 \bar{\xi}_n \leq \frac{z_n^6 \bar{\xi}_n^2}{\sqrt{z_n^6 \bar{\xi}_n^2 + \varrho_n^2}} + \varrho_n.$$

Then, one has

$$\begin{aligned}z_n^3 \left(-\frac{z_n^3 \bar{\xi}_n^2}{\sqrt{z_n^6 \bar{\xi}_n^2 + \varrho_n^2}} \right) &= -\frac{z_n^6 \bar{\xi}_n^2}{\sqrt{z_n^6 \bar{\xi}_n^2 + \varrho_n^2}} \leq -\frac{z_n^6 \bar{\xi}_n^2}{\sqrt{z_n^6 \bar{\xi}_n^2 + \varrho_n^2}} \\ &\leq \varrho_n - z_n^3 \bar{\xi}_n.\end{aligned}$$

According to Assumption 1, we know $\gamma \leq \mathfrak{h}'(v_\mu) \leq 1$. Keeping this in mind, it is easy to get

$$z_n^3 \mathfrak{h}'(v_\mu) \left(-\frac{z_n^3 \bar{\xi}_n^2}{\gamma \sqrt{z_n^6 \bar{\xi}_n^2 + \varrho_n^2}} \right) \leq -\frac{z_n^6 \bar{\xi}_n^2}{\sqrt{z_n^6 \bar{\xi}_n^2 + \varrho_n^2}} \leq \varrho_n - z_n^3 \bar{\xi}_n. \quad (60)$$

Thus, substituting (59) and (60) into (58), one can gain that

$$\begin{aligned}\mathcal{L}V_n &\leq -\frac{k_1}{4^\sigma} \sum_{j=1}^{n-1} z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^{n-1} z_j^{4\beta} - \sum_{j=1}^{n-1} \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 \\ &\quad - \frac{k_1}{4^\sigma} z_n^{4\sigma} - \frac{k_2}{4^\beta} z_n^{4\beta} + \frac{1}{2a_n^2} z_n^6 (\theta_n - \hat{\theta}_n - \tilde{\theta}_n) \psi_n^T \psi_n \\ &\quad + \frac{1}{4} \aleph^4 + \varrho_n + \frac{\tau_n^4}{4} + \frac{a_n^2}{2} + \frac{3}{4} \nu_n^2 + \frac{b_n}{\lambda_n} \hat{\theta}_n \tilde{\theta}_n + \sum_{j=1}^{n-1} \Delta_j \\ &\leq -\frac{k_1}{4^\sigma} \sum_{j=1}^n z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^n z_j^{4\beta} - \sum_{j=1}^{n-1} \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 + \sum_{j=1}^{n-1} \Delta_j \\ &\quad + \frac{1}{4} \aleph^4 + \varrho_n + \frac{\tau_n^4}{4} + \frac{a_n^2}{2} + \frac{3}{4} \nu_n^2 + \frac{b_n}{\lambda_n} \hat{\theta}_n \tilde{\theta}_n.\end{aligned}\quad (61)$$

Further, it's not difficult to gain

$$\frac{b_n}{\lambda_n} \tilde{\theta}_n \hat{\theta}_n = \frac{b_n}{\lambda_n} \tilde{\theta}_n (\theta_n - \tilde{\theta}_n) \leq \frac{b_n}{2\lambda_n} \theta_n^2 - \frac{b_n}{2\lambda_n} \tilde{\theta}_n^2. \quad (62)$$

Substituting (62) into (61) yields

$$\mathcal{L}V_n \leq -\frac{k_1}{4^\sigma} \sum_{j=1}^n z_j^{4\sigma} - \frac{k_2}{4^\beta} \sum_{j=1}^n z_j^{4\beta} - \sum_{j=1}^n \frac{b_j \tilde{\theta}_j^2}{2\lambda_j} + \sum_{j=1}^n \Delta_j, \quad (63)$$

where $\Delta_n = \frac{3}{4} \nu_n^2 + \frac{a_n^2}{2} + \frac{\tau_n^4}{4} + \frac{b_n}{2\lambda_n} \theta_n^2 + \frac{1}{4} \aleph^4 + \varrho_n$ is a constant. By using Lemmas 4 and 6, one can get that

$$-\frac{k_1}{4^\sigma} \sum_{j=1}^n z_j^{4\sigma} \leq -k_1 \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\sigma, \quad (64)$$

$$-\frac{k_2}{4^\beta} \sum_{j=1}^n z_j^{4\beta} \leq -n^{1-\beta} k_2 \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\beta, \quad (65)$$

$$\begin{aligned} \left(\sum_{j=1}^n \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 \right)^\sigma &\leq \frac{(\sigma^\frac{1}{\sigma})^\sigma}{\sigma} \left[\left(\sum_{j=1}^n \frac{b_j}{2\lambda_1} \tilde{\theta}_j^2 \right)^\sigma \right]^\frac{1}{\sigma} \\ &+ (1^{1-\sigma})^\frac{1}{\sigma} \frac{1-\sigma}{(\sigma^\sigma)^{\frac{1}{1-\sigma}}} = \sum_{j=1}^n \frac{b_j}{2\lambda_1} \tilde{\theta}_j^2 + \rho_1, \end{aligned} \quad (66)$$

$$\begin{aligned} -\sum_{j=1}^n b_j \left(\frac{1}{2\lambda_j} \tilde{\theta}_j^2 \right)^\beta &\leq -b \sum_{j=1}^n \left(\frac{1}{2\lambda_j} \tilde{\theta}_j^2 \right)^\beta \\ &\leq -n^{1-\beta} b \left(\sum_{j=1}^n \frac{1}{2\lambda_1} \tilde{\theta}_j^2 \right)^\beta, \end{aligned} \quad (67)$$

where $\rho_1 = (1-\sigma)\sigma^{\frac{\sigma}{1-\sigma}}, b = \min_{1 \leq j \leq n} \{b_j\}$.

From (61), we have

$$\begin{aligned} \mathcal{L}V_n(Z(t)) &\leq -k_1 \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\sigma - \frac{1}{2} \left(\sum_{j=1}^n \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 \right)^\sigma \\ &- n^{1-\beta} k_2 \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\beta + \frac{1}{2} \left(\sum_{j=1}^n \frac{b_j \tilde{\theta}_j^2}{2\lambda_j} \right)^\sigma \\ &+ \sum_{j=1}^n \Delta_j + \sum_{j=1}^n b_j \left(\frac{\tilde{\theta}_j^2}{4\lambda_j} \right)^\beta - \sum_{j=1}^n b_j \left(\frac{\tilde{\theta}_j^2}{4\lambda_j} \right)^\sigma \\ &- \sum_{j=1}^n \frac{b_j \tilde{\theta}_j^2}{4\lambda_j}. \end{aligned} \quad (68)$$

Substituting (64), (65), (66) and (67) into (68), we can gain

$$\begin{aligned} \mathcal{L}V_n(Z(t)) &\leq -k_1 \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\sigma - \frac{1}{2} \left(\sum_{j=1}^n \frac{b_j}{2\lambda_j} \tilde{\theta}_j^2 \right)^\sigma \\ &- n^{1-\beta} k_2 \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\beta - n^{1-\beta} b \left(\sum_{j=1}^n \frac{1}{2\lambda_j} \tilde{\theta}_j^2 \right)^\beta \\ &+ \sum_{j=1}^n \Delta_j + \rho_1 + \sum_{j=1}^n b_j \left(\frac{\tilde{\theta}_j^2}{4\lambda_j} \right)^\beta - \sum_{j=1}^n \frac{b_j \tilde{\theta}_j^2}{4\lambda_j} \\ &\leq -\mu_1 \left\{ \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\sigma + \left(\sum_{j=1}^n \frac{\tilde{\theta}_j^2}{2\lambda_j} \right)^\sigma \right\} \\ &- \bar{\mu}_2 \left\{ \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\beta + \left(\sum_{j=1}^n \frac{\tilde{\theta}_j^2}{2\lambda_j} \right)^\beta \right\} \\ &+ \sum_{j=1}^n \Delta_j + \rho_1 + \sum_{j=1}^n b_j \left(\frac{\tilde{\theta}_j^2}{4\lambda_j} \right)^\beta - \sum_{j=1}^n \frac{b_j \tilde{\theta}_j^2}{4\lambda_j}, \end{aligned} \quad (69)$$

where $\mu_1 = \min\{k_1, \frac{b^\sigma}{2}\}, \bar{\mu}_2 = \min\{n^{1-\beta} k_2, 2^{-\beta} n^{1-\beta} b\}$.

Then, according to lemma 4, one has

$$\begin{aligned} V_n^\sigma(Z(t)) &\leq \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\sigma + \left(\sum_{j=1}^n \frac{\tilde{\theta}_j^2}{2\lambda_j} \right)^\sigma, \\ V_n^\beta(Z(t)) &\leq n^{\beta-1} \left\{ \left(\sum_{j=1}^n \frac{z_j^4}{4} \right)^\beta + \left(\sum_{j=1}^n \frac{\tilde{\theta}_j^2}{2\lambda_j} \right)^\beta \right\}. \end{aligned} \quad (70)$$

Finally, combining (70) with (69), one gets

$$\begin{aligned} \mathcal{L}V_n(Z(t)) &\leq -\mu_1 (V_n(Z(t)))^\sigma - \mu_2 (V_n(Z(t)))^\beta \\ &+ \sum_{j=1}^n \Delta_j + \rho_1 + \sum_{j=1}^n b_j \left(\frac{\tilde{\theta}_j^2}{4\lambda_j} \right)^\beta - \sum_{j=1}^n \frac{b_j \tilde{\theta}_j^2}{4\lambda_j}, \end{aligned} \quad (71)$$

where $\mu_2 = n^{1-\beta} \bar{\mu}_2$. Now, we discuss the estimations of the last two terms in the right side of equation (71). Suppose that there exists an unknown constant m_j satisfying $|\tilde{\theta}_j| < m_j$, then we will analyze through two situations.

Case 1: If $m_j < 2\sqrt{\lambda_j}$, then $\frac{\tilde{\theta}_j^2}{4\lambda_j} < 1$. Since $\beta > 1$, one can get $\sum_{j=1}^n b_j \left(\frac{\tilde{\theta}_j^2}{4\lambda_j} \right)^\beta - \sum_{j=1}^n \frac{b_j \tilde{\theta}_j^2}{4\lambda_j} < 0$.

So it can be inferred that

$$\mathcal{L}V_n(Z(t)) \leq -\mu_1 (V_n(Z(t)))^\sigma - \mu_2 (V_n(Z(t)))^\beta + \rho_2, \quad (72)$$

where $\rho_2 = \sum_{j=1}^n \Delta_j + \rho_1$.

Case 2: If $m_j \geq 2\sqrt{\lambda_j}$, then $\frac{\tilde{\theta}_j^2}{4\lambda_j} < 1$ or $\frac{\tilde{\theta}_j^2}{4\lambda_j} > 1$.

However, we can drive that the maximum value of the objective function $F(\chi) = \sum_{j=1}^n b_j \chi_j^\beta - \sum_{j=1}^n b_j \chi_j$ is

$$\sum_{j=1}^n b_j \left(\frac{m_j^2}{4\lambda_j} \right)^\beta - \sum_{j=1}^n \frac{b_j m_j^2}{4\lambda_j}.$$

Therefore, for $\chi_j = \frac{\tilde{\theta}_j^2}{4\lambda_j}$, one has the following equation holds $\sum_{j=1}^n b_j \left(\frac{\tilde{\theta}_j^2}{4\lambda_j} \right)^\beta - \sum_{j=1}^n \frac{b_j \tilde{\theta}_j^2}{4\lambda_j} \leq \sum_{j=1}^n b_j \left(\frac{m_j^2}{4\lambda_j} \right)^\beta - \sum_{j=1}^n \frac{b_j m_j^2}{4\lambda_j}$.

Thus, we can easily conclude

$$\mathcal{L}V_n(Z(t)) \leq -\mu_1 (V_n(Z(t)))^\sigma - \mu_2 (V_n(Z(t)))^\beta + \rho_3, \quad (73)$$

where $\rho_3 = \sum_{j=1}^n b_j \left(\frac{m_j^2}{4\lambda_j} \right)^\beta - \sum_{j=1}^n \frac{b_j m_j^2}{4\lambda_j} + \rho_2$.

Based on Cases 1 and 2, it can be inferred that

$$\mathcal{L}V_n(Z(t)) \leq -\mu_1(V_n(Z(t)))^\sigma - \mu_2(V_n(Z(t)))^\beta + \rho, \quad (74)$$

where

$$\rho = \begin{cases} \rho_2, & m_j < 2\sqrt{\lambda_j}. \\ \rho_3, & m_j \geq 2\sqrt{\lambda_j}. \end{cases}$$

In accordance with Theorem 1, one obtains that the discussed system (26) is practical fixed-time stable in probability, and the settling time T_p satisfies

$ET_p \leq T_{\max} := \frac{\Gamma(\frac{1-\sigma}{\beta-\sigma})\Gamma(\frac{\beta-1}{\beta-\sigma})}{(2^\sigma-1)\mu_1(\beta-\sigma)} \left[\frac{(2^\sigma-1)\mu_1}{\mu_2} \right]^{\frac{1-\sigma}{\beta-\sigma}}$. In other words, $V_n(Z(t))$ and $Z(t)$ are fixed-time bounded almost surely, and $Z(t)$ converges to the following compact set in probability when $t \geq T_p$

$$\Omega = \left\{ Z(t) \mid E[V_n(Z(t))] \leq \min \left\{ \frac{2\rho}{\mu_1}, \frac{2\rho}{\mu_2} \right\} \right\}. \quad (75)$$

In view of the definition of $V_n(Z(t))$ and $Z(t)$, we can attain the boundness of z_i and $\hat{\theta}_i$ for $\forall i = 1, \dots, n$. In light of $\tilde{\theta}_i = \theta_i - \hat{\theta}_i$, one can infer that $\hat{\theta}_i$ is bounded. Due to the relation between ζ_{i+1} and $(z_i, \hat{\theta}_i)$, it can be inferred that ζ_{i+1} is also bounded. Thus, it can be concluded the boundness of x_i . Ultimately, we can deduce that all the variables are bounded almost surely. That is, the original considered system (1) is practical fixed-time stable in probability.

Till now, the proof of Theorem 2 is finished.

Remark 6. It can be seen from equation (71) that the control performance depends on the values of the constants μ_1 and μ_2 , which are determined by the primary parameters, that is, $a_1, \dots, a_n, k_1, k_2, \lambda_1, \dots, \lambda_n, b_1, \dots, b_n$. However, the altitude of the controller u may become larger, which is wanted to be avoided. Consequently, in practical applications, the design parameters should be carefully adjusted by considering the tradeoff between the control performance and control action.

Simulation

In this section, the simulation results of two examples will further demonstrate the conclusion of Theorem 2.

Example 1:

$$\begin{cases} dx_1 = x_2 dt + 0.2x_1^2 \sin(x_1) d\omega, \\ dx_2 = (u(v) + 0.1(x_1 + x_2)^2) dt + \frac{3x_1}{20} \cos(x_2) d\omega, \end{cases} \quad (76)$$

where $u(v)$ is the system input control with saturation like as:

$$u = \text{sat}(v) = \begin{cases} 5, & v \geq 5. \\ v, & |v| < 5. \\ -5, & v \leq -5. \end{cases}$$

To use the approximated method of FLSs, we select the following fuzzy basic functions

$$\begin{aligned} \tilde{\psi}_{A'_1}(x_1) &= \exp^{-0.5(x_1 + l)^2}, \\ \tilde{\psi}_{A'_2}(x_2) &= \exp^{-0.5(x_2 + l)^2}, \\ \tilde{\psi}_{A'_3}(\hat{\theta}_1) &= \exp^{-0.5(\hat{\theta}_1 + l)^2}, \quad l = -5, -3, -1, 0, 1, 3, 5. \end{aligned} \quad (77)$$

In view of previous devised proceeding, one can conclude that

$$\begin{cases} \bar{\xi}_1 = \frac{k_1}{4^\sigma} z_1^{4\sigma-3}, \\ \bar{\xi}_2 = \frac{k_1}{4^\sigma} z_2^{4\sigma-3}, \\ \zeta_2 = -\frac{z_1^3 \bar{\xi}_1^2}{\sqrt{z_1^6 \bar{\xi}_1^2 + q_1^2}} - \frac{1}{2a_1^2} \hat{\theta}_1 z_1^3 \|\psi_1\|^2 - \frac{k_2}{4^\beta} z_1^{4\beta-3} - \frac{3}{2} z_2, \\ v = -\frac{1}{\gamma} \left(\frac{z_2^3 \bar{\xi}_2^2}{\sqrt{z_2^6 \bar{\xi}_2^2 + q_2^2}} + \frac{1}{2a_2^2} \hat{\theta}_2 z_2^3 \|\psi_2\|^2 + \frac{k_2}{4^\beta} z_2^{4\beta-3} + \frac{7}{4} z_n \right), \\ \hat{\theta}_1 = \frac{\lambda_1}{2a_1^2} z_1^6 \|\psi_1\|^2 - b_1 \hat{\theta}_1, \\ \hat{\theta}_2 = \frac{\lambda_2}{2a_2^2} z_2^6 \|\psi_2\|^2 - b_2 \hat{\theta}_2. \end{cases} \quad (78)$$

By trying and calculating by Matlab, the parameters are chosen as follows: $a_1 = 2, a_2 = 4, k_1 = 2, k_2 = 1, \sigma = 0.75, \beta = 1.75, \lambda_1 = 8, \lambda_2 = 12, b_1 = 2, b_2 = 3, q_1 = 0.095, q_2 = 0.095, \gamma = \frac{1}{e^4}$. According to Theorem 1, it can be calculated that the upper estimation of the settling time $T_{\max} = 8.7449s$. Then, the two initial values of x are selected as $x_0 = (0.8, 0.8)^T$ and $x_0 = (-1.8, 5)^T$, and the initial value of $\hat{\theta}$ is taken as $\hat{\theta}_0 = (-1, 1)^T$.

The simulation results of system variables with $x_0 = (0.8, 0.8)^T$ and $x_0 = (-1.8, 5)^T$ are shown in Figures 3 to 10. Observing the profiles of states $x_i (i = 1, 2)$ for two different initial conditions in Figures 3 and 7, one obtains that the system states can converge to a small region of containing zero point when $t > 5.2s$ under the actual control $u(v)$.

On the other hands, Figures 4, 5, 8 and 9 display the responding curves of the control input v and the saturated actual control $u(v)$. From these figures, we can find that the control input v and $u(v)$ also can converge within 5.2s, which implies that the designed controller effectively compensates for the influence of input saturation on the system. As we can see in Figures 6 and 10, the adaptive laws can arriving at stable values

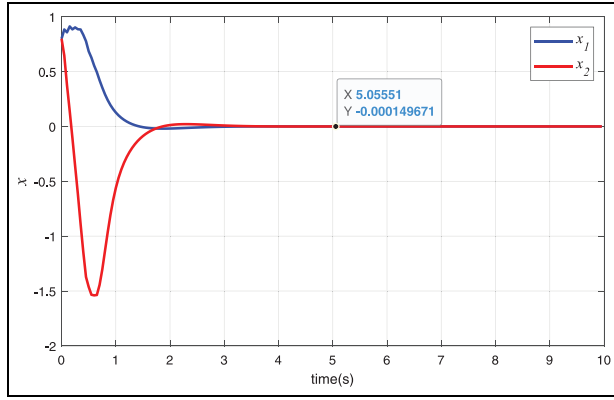


Figure 3. System state $x(t)$ with $x_0 = (0.8, 0.8)^T$ of Example 1.

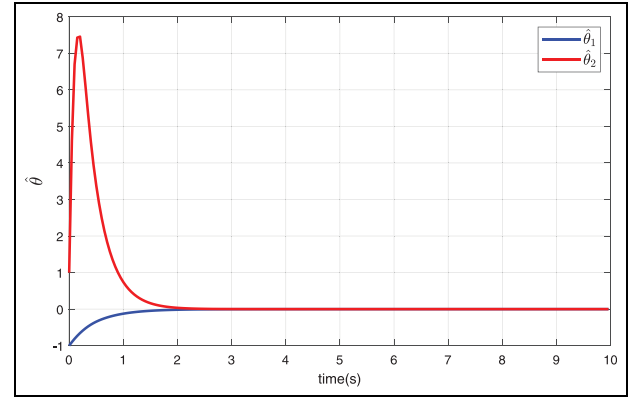


Figure 6. Adaptive laws with $x_0 = (0.8, 0.8)^T$, $\hat{\theta}_0 = (-1, 1)^T$ of Example 1.

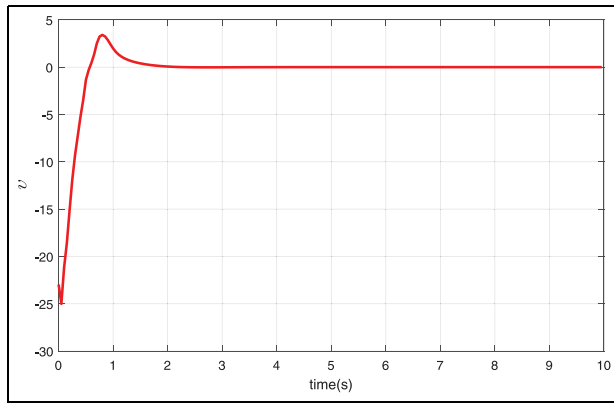


Figure 4. Control input v with $x_0 = (0.8, 0.8)^T$ of Example 1.

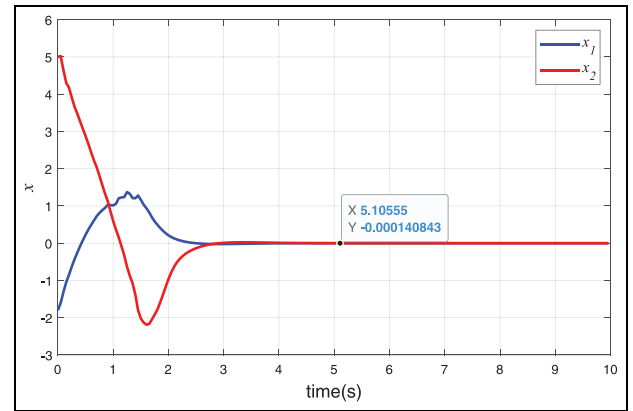


Figure 7. System state $x(t)$ with $x_0 = (-1.8, 5)^T$ of Example 1.

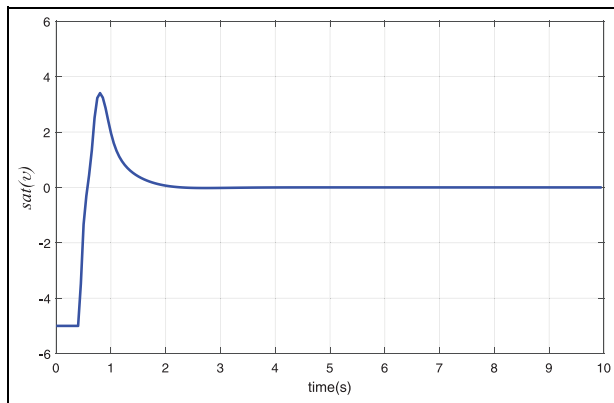


Figure 5. Saturation control u with $x_0 = (0.8, 0.8)^T$ of Example 1.

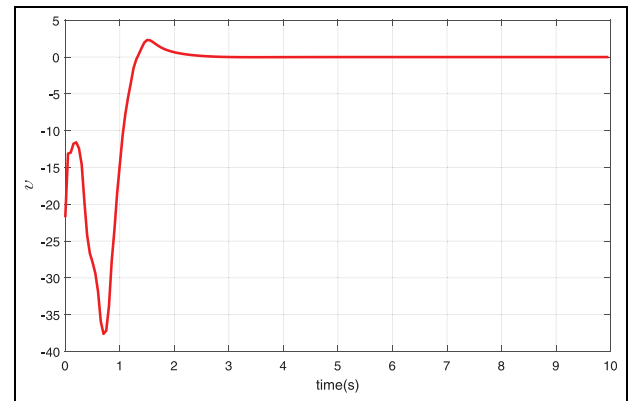


Figure 8. Control input v with $x_0 = (-1.8, 5)^T$ of Example 1.

within $5.2s$, too. All of above indicate that the settling time $T_p \doteq 5.2s$ and satisfies $T_p \leq T_{max} = 8.7449s$. In other words, all the variables are bounded within $8.7449s$.

Moreover, we notice that controller (78) would become a finite-time stabilizer if $k_2 = 0$. To compare fixed-time stability with finite-time stability, we also present Figures 11 and 12 to show the simulation result

of system (76) under the finite-time stabilizing controller. From the figures, it can be observed that the finite-time stabilizing controller is valid, too. But the settling time for the two different initial conditions are respectively $\hat{T}_p \doteq 7.7s$ and $\hat{T}_p \doteq 9.3s$, the upper bound estimations correspondingly become larger. In other words, the fixed-time stabilizing controller not only can

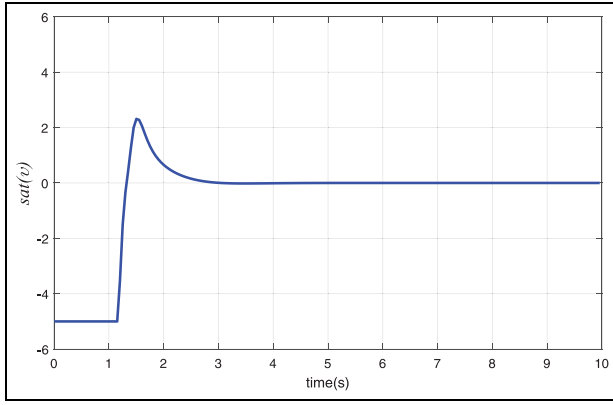


Figure 9. Saturation control u with $x_0 = (-1.8, 5)^T$ of Example 1.

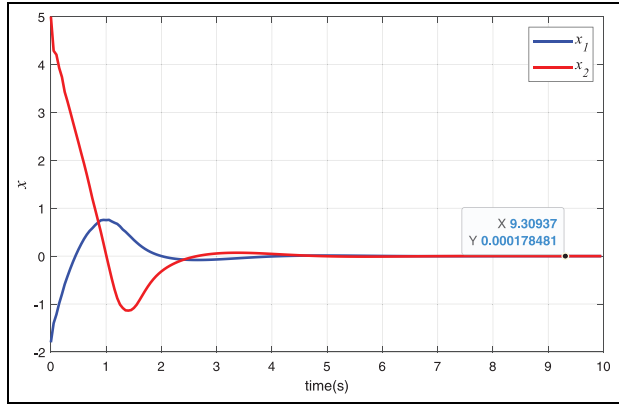


Figure 12. System state $x(t)$ with $x_0 = (-1.8, 5)^T$ of Example 1 under finite-time controller.

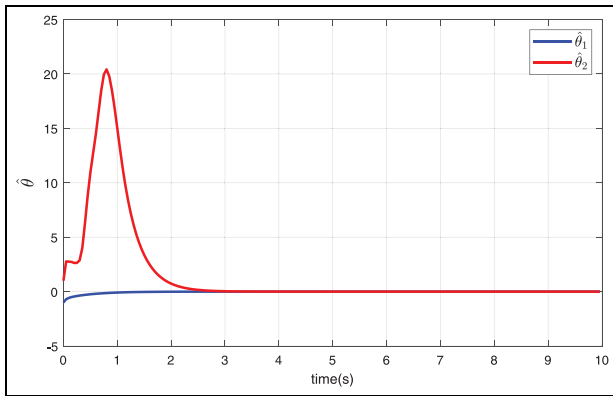


Figure 10. Adaptive laws with $x_0 = (-1.8, 5)^T$, $\hat{\theta} = (-1, 1)^T$ of Example 1.

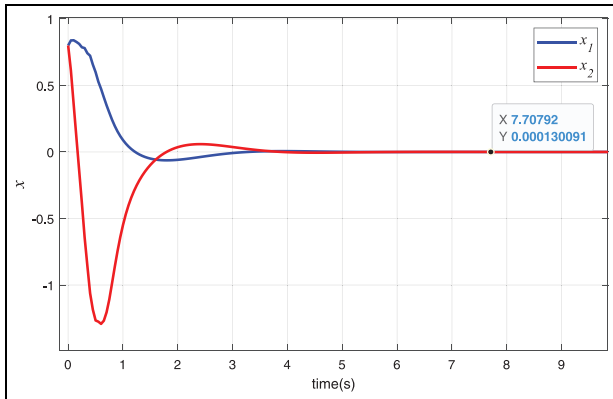


Figure 11. System state $x(t)$ with $x_0 = (0.8, 0.8)^T$ of Example 1 under finite-time controller.

guarantee that the upper bound estimation of the settling time is independent of initial conditions, but also can make the system states converge faster than the system controlled by the finite-time one. Thus, the simulation results illustrate the effectiveness of the proposed scheme.

Example 2 : The dynamic of one-link manipulator⁴¹ is discussed to demonstrate the presented control program. The model of the system with stochastic disturbance is denoted as

$$\begin{cases} B\ddot{q} + J\dot{q} + L\sin(q) = \varsigma + \varsigma_d, \\ K\varsigma + P\varsigma + H_m\dot{q} = u(v), \end{cases} \quad (79)$$

where $\ddot{q}$, $\dot{q}$, and q represent the acceleration, velocity, and link position, respectively; ς is the torque; $\varsigma_d = 0.1\sin(q^2)\frac{d\omega}{dt}$ with ω denoting stochastic disturbance defined as in system (1). Additionally, $u(v) = \text{sat}(v)$ is the input control electromechanical torque with a saturation limit of $5N \cdot m$. The relative parameters are $B = 1Kg \cdot m^2$, $J = 1Nm \cdot s/rad$, $L = 2$, $K = 1H$, $P = 1\Omega$, $H_m = 2Nm/A$. Defining $x_1 = q$, $x_2 = \dot{q}$, $x_3 = \varsigma$, then system (79) can be expressed as

$$\begin{cases} dx_1 = x_2 dt, \\ dx_2 = (x_3 - x_2 - 2\sin(x_1))dt + 0.1\sin(x_1^2)d\omega, \\ dx_3 = (u(v) - 2x_2 - x_3)dt, \end{cases} \quad (80)$$

where the saturation data is chosen as $u_M = 5$. For designing the controller, the similar fuzzy basic functions applied in Example 1 are employed in Example 2. The design parameters are chosen as $\sigma = 0.95$, $\beta = 2$, $a_1 = 2$, $a_2 = 4$, $a_3 = 5$, $k_1 = 2$, $k_2 = 1$, $\lambda_1 = 8$, $\lambda_2 = 12$, $\lambda_3 = 15$, $b_1 = 2$, $b_2 = 3$, $b_3 = 4$, $q_1 = 0.05$, $q_2 = 0.05$, $q_3 = 0.05$, $\gamma = \frac{1}{e^4}$.

For the initial vectors $x = (0.8, -1, 1)^T$ and $\hat{\theta} = (-0.2, 0.2, 0.3)^T$, the simulation results are revealed in Figures 13 to 16. In light of Theorem 1, it can be inferred that $T_{\max} = 25.4615s$. From Figure 13, we are able to conclude that the state vector converges to a small region of the origin when $t \geq T_p \doteq 5.5s \leq 25.4615s$. Figures 14 and 15 represent the curves of input control and saturation control input. From these figures, we also have the conclusion that the control

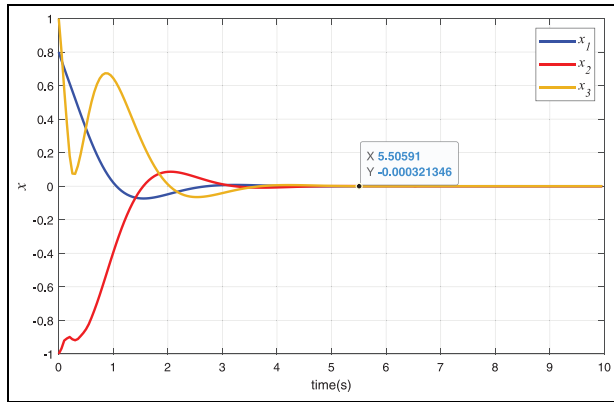


Figure 13. System state $x(t)$ with $x_0 = (0.8, -1, 1)^T$ of Example 2.

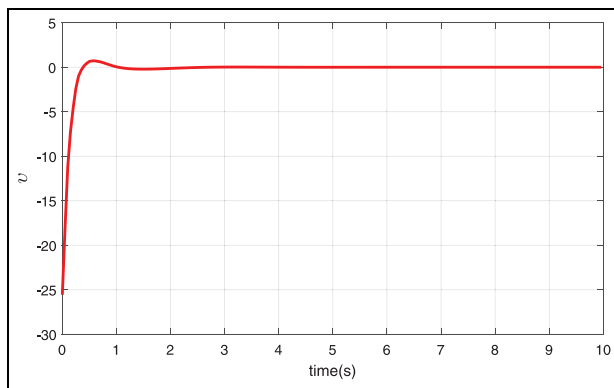


Figure 14. Control input v of Example 2.

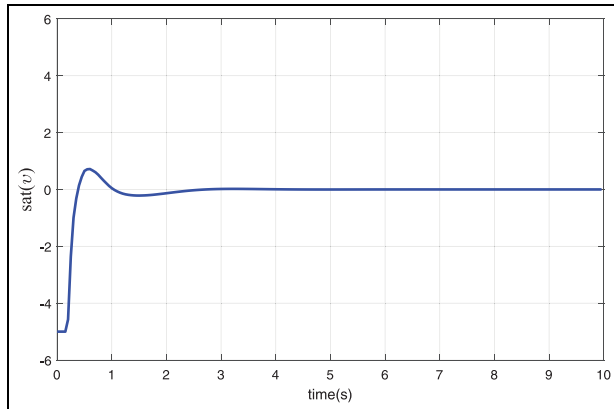


Figure 15. Saturation control $u = \text{sat}(v)$ of Example 2.

input satisfy the saturation restriction and converge to a small region of zero when $t \geq T_p \doteq 5.5s$. Figure 16 confirms that the adaptive laws can reach the steady state within 5.5s.

According to above results, it follows that all the variables of the system are bounded and converge to a set containing equilibrium point within a fixed time.

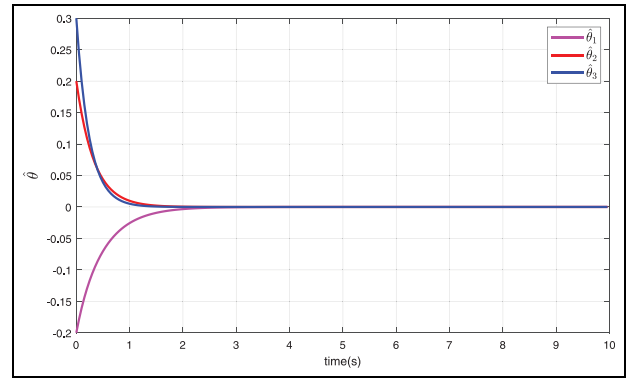


Figure 16. Adaptive laws with $\hat{\theta}_0 = (-0.2, 0.2, 0.3)^T$ of Example 2.

Hence, the suggested control strategy can be utilized to resolve the fixed-time control issue of stochastic systems with input saturation.

Conclusion

In this note, the adaptive fuzzy fixed-time control problem is addressed for stochastic systems with input saturation. To deal with input saturation and unknown functions, Gaussian error function and FLSs are applied in the controller design process. Furthermore, a stability criterion and a fuzzy adaptive fixed-time control strategy are proposed to ensure that all variables of the discussed system are bounded. Eventually, the validity of the developed control scheme is demonstrated through numerical simulations. However, there are still some limitations of the proposed control strategy. For examples, we haven't take into account disturbances and singularity problem, and the proposed controller is a bit complex to compute, which may make not easy for real-time implementation or hardware-in-the-loop simulation. In the future, we aim to develop fixed-time controller design framework with low complexity avoiding the singularity problem and investigate more general models of stochastic nonlinear systems, such as those featuring disturbances and time-varying delays. Additionally, our future endeavors include exploring the tracking control, the integration of fixed-time control theory with other advance control methodologies to enhance the control performance of systems.

ORCID iD

Liandi Fang  <https://orcid.org/0000-0001-8347-791X>

Ethical considerations

This article does not contain any studies with human or animal participants. There are no human participants in this article and informed consent is not required.

Consent to participate

The authors declared no potential conflicts of interest with respect to the research and the order of authorship of this article.

Consent for publication statement

All authors have given consent for the manuscript to be published in its current form.

Author Contributions

Liandi Fang: Conceptualization; Methodology; Writing—original draft; Zhenzhen Long and Daohong Zhu: Validation; Writing—review and editing.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work is supported by the National Natural Science Foundation of China (62103306), Scientific Research Project of Anhui Higher Education Institutions (2022AH020094 and 2023AH051661), Talent Research Launch Fund Project of Tongling University (2023tlxyc40), and Graduate Research Innovation Fund Project of Tongling University (23tlcx11).

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

No data were used to support this study.

Trial registration number/date

This article does not contain any trial.

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